Lecture 12 // Oct. 1 // The Chern-Simons functional & Reducibles

Last time: We defined the Chern-Simons functional

$$CS(A) = \frac{1}{8\pi^2}\int_Y \mathrm{tr}\left(A\wedge dA + \frac{2}{3}A\wedge A\wedge A\right)$$

and saw that $\mathrm{grad}_A CS = *F_A$

We also interpreted $\mathrm{crit}(CS)$ in two ways:

$$\mathrm{Crit}(CS) = \left\{\begin{array}{l}\text{flat connections}\\ \text{modulo gauge}\end{array}\right\} \leftrightarrow \mathcal{R}$$

Today, we'll complete the pf of thm from last time

Thm Critical Points of the Chern-Simons functional modulo gauge are in bijection with $SU(2)$ representations of $\pi_1(Y)$ modulo conjugation.

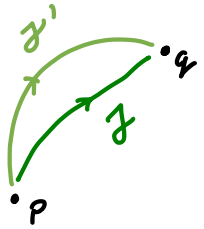
Pf (cont)

We must show injectivity.

Let ∇_0, ∇_1 be two connections ST $\mathrm{hol}_p|^{\nabla_0}$ and $\mathrm{hol}_p^{\nabla_1}$ are related by conjugation via some fixed $g_p \in SU(2)$

We want to show ∇_0 and ∇_1 are gauge equivalent. We construct a global gauge transformation from g_p as follows: For any q, choose a path γ from p to q. Then define

$$g_q = \mathrm{hol}^{\nabla_1}(\gamma)\circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})$$



Is g_q well-defined?

Yes: Choose another γ' from p to q.

Then $g'_q = \mathrm{hol}^{\nabla_1}(\gamma')\circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')$

and
$$\begin{aligned} g_q g_q'^{-1} &= \left(\mathrm{hol}^{\nabla_1}(\gamma)\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\right)\circ\left(\mathrm{hol}^{\nabla_1}(\gamma')\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')\right)^{-1}\\ &= \left(\mathrm{hol}^{\nabla_1}(\gamma)\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\right)\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')^{-1}\circ g_p^{-1}\circ \mathrm{hol}^{\nabla_1}(\gamma')^{-1}\\ &= \mathrm{hol}^{\nabla_1}(\gamma)\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\circ \mathrm{hol}^{\nabla_0}(\gamma')\circ g_p^{-1}\circ \mathrm{hol}^{\nabla_1}(\bar{\gamma}') \end{aligned}$$

Since $\text{hol}^{\nabla_0}(\bar{\gamma}) \circ \text{hol}^{\nabla_0}(\gamma') \in \text{hol}_p^{\nabla_0}$,

and $\text{hol}_p^{\nabla_0}$ & $\text{hol}_p^{\nabla_1}$ are related by $g_p \in SU(2)$,

$$g_p \circ \text{hol}^{\nabla_0}(\bar{\gamma}) \circ \text{hol}^{\nabla_0}(\gamma') \circ g_p^{-1} = \text{hol}^{\nabla_1}(\bar{\gamma}) \circ \text{hol}^{\nabla_1}(\gamma'),$$

so

$$g_q g_p^{-1} = \text{hol}^{\nabla_1}(\gamma) \circ \text{hol}^{\nabla_1}(\bar{\gamma}) \circ \text{hol}^{\nabla_1}(\gamma') \circ \text{hol}^{\nabla_1}(\bar{\gamma}') = id.$$

By construction, $\text{hol}^{\nabla_0} = g_q^{-1} \circ \text{hol}^{\nabla}(\gamma) \circ g_p$, so certainly the parallel transports of ∇_0, ∇_1 behave as if related by a gauge transformation. □

The Instanton Chain Complex

Recall that the generators of the ICC are the crit pts of the CS functional, and the differential is defined by looking at the grad. flow of CS, $\text{grad}_A(CS) = *F_A$. But how can we do this on a three manifold? The general slogan is that studying the grad. flow of CS on a three manifold Y^3 is the same as studying SD/ASD YM connections on $Y^3 \times \mathbb{R}$. There's a very precise connection btwn the three and four-manifold stories; this slogan is the first piece. Space Instead of studying SD/ASD on closed X^4, we looked at the open 4-mfd $\mathbb{R} \times Y$. On $\mathbb{R} \times Y$, consider the triv. $SU(2)$-bundle. $\mathbb{R} \times Y \times \mathbb{C}^2$. Since $c_2(\mathbb{R} \times Y) = 0$. By the Künneth formula, $c_2(\mathbb{R} \times Y) = \sum_{i+j=2} c_i(\mathbb{R}) \otimes c_j(Y) = 0$, this is only one.

Now, lets fix a metric g_Y on Y, taking the product metric $dt^2 + g_Y$ on $\mathbb{R} \times Y$. We may use this to orient $\mathbb{R} \times Y$ by declaring a positively-oriented basis to be $\{\underbrace{dt}_{\mathbb{R}}, \underbrace{dx, dy, dz}_{Y \text{ directions}}\}$.

Let's think about the bundle as a 1P1 of objects over Y

A GT g over over $\mathbb{R}\times Y$ is 1PF of GT's g_t:

eg, a GT on Y sm. varying with t

A cxn (rather, a cxn 1-form) on $\mathbb{R}\times Y$ is, in general, of the form $A = A_t + B_t\,dt$. This is achieved by separating the dt from dx, dy, dz in 1-form so

A_t consists of one forms in Y and B_t is a matrix of 0-forms.

Note that A_t, B_t still vary in t, since the x, y, z coordinates are themselves fxns of t: $x = x(t)$, $y = y(t)$, $z = z(t)$. Also note that B_t is not a cxn on Y.

For each t, define the inclusion map $\iota_t : Y \to \{t\}\times Y \subset \mathbb{R}\times Y$; then A_t is $\iota_t^*(A)$. It will be convenient for us to define the following.

Def A is in temporal gauge if $B_t = 0$.

Lemma Up to GT on $\mathbb{R}\times Y$, every A can be put into temporal gauge.

Pf We want to find a 1PF of GTs $\{g_t\}$ which kills the dt component.

Applying the transformation law, g_t sends B_t to $g_t^{-1} B_t g_t^{-1} - g_t^{-1} dg_t$. Therefore g_t acts on A via

$$A \xmapsto{\curvearrowright g_t} g_t^{-1} A g_t + g_t^{-1} dg_t$$

Simplifying,

$$g_t^{-1} A g_t + g_t^{-1} dg_t = g_t^{-1}(A_t + B_t\,dt) g_t + g_t^{-1} dg_t$$
$$= g_t^{-1} A_t g_t + g_t^{-1} B_t g_t\,dt + g_t^{-1} dg_t$$

The dt comp. of $g_t^{-1} dg_t$ is $g_t^{-1} \frac{\partial}{\partial t} g_t\,dt$, so the dt comp. of $g_t^{-1} A g_t + g_t^{-1} dg_t$ is

$$g_t^{-1} B_t g_t\,dt + g_t^{-1} \frac{\partial}{\partial t} g_t\,dt.$$

Therefore we would like to show that for every t, there is

a g_t such that

$$g_t^{-1} B_t g_t \, dt + g_t^{-1} \frac{\partial}{\partial t} g_t \, dt = 0$$

In other words, we want to solve the (IPF) of ODEs:

$$g_t^{-1} B_t g_t \, dt + g_t^{-1} \frac{\partial}{\partial t} g_t \, dt = 0$$

$$\Leftrightarrow \quad B_t g_t \, dt + \frac{\partial}{\partial t} g_t \, dt = 0$$

$$\Leftrightarrow \quad B_t g_t + \frac{\partial}{\partial t} g_t = 0$$

$$\Leftrightarrow \quad \frac{\partial}{\partial t} g_t = -B_t g_t$$

For each fixed $y \in Y$, we want that

$$\frac{\partial}{\partial t} g_t(y) = -B_t(y) \, g_t(y).$$

Let's interpret the right hand side.

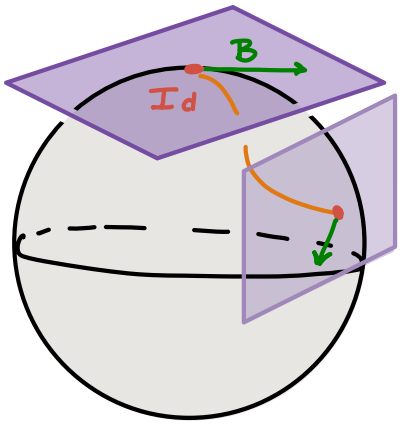
Note that $g_t(y)$ is a path in $SU(2)$, and $B_t(y)$ is a time-varying element of the lie algebra of $SU(2)$, $\mathfrak{su}(2)$, which is the tangent space at the identity $T_{Id} SU(2)$.

Explicitly, for a fixed t and y, $B_t(y)$ is a 2×2 matrix in $\mathfrak{su}(2)$, and $g_t(y)$ is a 2×2 matrix in $SU(2)$.

We can get a new 2×2 matrix in $SU(2)$ by right-multiplying $B_t(y)$ by $g_t(y)$:

$$B_t(y) \cdot g_t(y) = \begin{bmatrix} b_{11}(t,y) & b_{21}(t,y) \\ b_{12}(t,y) & b_{22}(t,y) \end{bmatrix} \cdot \begin{bmatrix} g_{11}(t,y) & g_{21}(t,y) \\ g_{12}(t,y) & g_{22}(t,y) \end{bmatrix}$$

$T_{Id} SU(2) \cong \mathfrak{su}(2)$



Doing so at each t, still fixing y, gives us the pushfoward of $B_t(y)$ under right mult. by $g_t(y)$, defining a v.f. over all of $SU(2)$.

Combining this with the LHS, if this vector field equals $\frac{d}{dt} g_t(y)$, the path must be an integral curve (eg, a flowline) for this VF. By ODE theory, this always exists; if the initial conditions are fixed, it is unique. □

up to GT, we can think of D on $\mathbb{R}_{xy}$ as $\{\nabla_t\}$ on Y

We're now able to ask: what do the SDYM equations look like?
First, the self-duality equation is $F_A = *F_A$.
Let $A = A_t$ be in temporal gauge. Calculating $F_A = dA_t + A \wedge A$,
note that it's not true that the 4-dimensional curvature is a 1PF 3-dimensional curvatures. Even though the connection itself has no dt component, the entries of this matrix are functions in all four variables, ie, these entries are allowed to vary in t. This means that dA will have a $\frac{\partial}{\partial t} A_t \, dt$ component, which encodes how the curvature of Y changes in the time direction.

$$dA = dt \wedge \frac{\partial}{\partial t} A_t + \left(\sum_{i=1}^{3} dx_i \wedge \frac{\partial}{\partial x_i} A_t \right)$$

$$= \frac{-\partial}{\partial t} A_t \wedge dt + \underbrace{\left(\sum_{i=1}^{3} dx_i \wedge \frac{\partial}{\partial x_i} A_t \right)}_{d^{(3)} A_t}$$

$$\Rightarrow F_A = \frac{-\partial}{\partial t} A_t \wedge dt + d^{(3)} A_t + A_t \wedge A_t$$

For notational convenience, define $F_{A_t} := d^{(3)} A_t + A_t \wedge A_t$
Then $F_A = -\frac{\partial}{\partial t} A_t \wedge dt + F_{A_t}$.

To calculate $*F_A$, lets find the relationship between the three and four-dimensional Hodge Stars. There are two main cases: if a 2-form has a time component, and if it doesnt.

1) $*^{(4)}(dx_1 \wedge dt) = -dx_2 \wedge dx_3 = -*^{(3)} dx_1$

2) $*^{(4)}(dx_1 \wedge dx_2) = -dx_3 \wedge dt = -*^{(3)}(dx_1 \wedge dx_2) \wedge dt$

This generalizes to the following relationship: let $i,j,k \in \{1,2,3\}$.

1) $*^{(4)}(dx_i \wedge dt) = -dx_j \wedge dx_k = -*^{(3)} dx_i$

2) $*^{(4)}(dx_i \wedge dx_j) = -dx_k \wedge dt = -*^{(3)}(dx_i \wedge dx_j) \wedge dt$

Thus $*^{(4)} F_A$ can be expressed using $*^{(3)}$ as

$$*^{(4)} F_A = *^{(4)}\left(\frac{-\partial}{\partial t} A_t \wedge dt + F_{A_t}\right)$$

$$= *^{(4)}\left(\frac{-\partial}{\partial t} A_t \wedge dt\right) + *^{(4)} F_{A_t}$$

$$= *^{(3)} \frac{\partial}{\partial t} A_t - *^{(3)} F_{A_t} \wedge dt$$

In full, the SDYM equation becomes:

$$F_A = *F_A.$$

$$-\frac{\partial}{\partial t} A_t \wedge dt + F_{A_t} = *^{(3)} \frac{\partial}{\partial t} A_t - *^{(3)} F_{A_t} \wedge dt$$

$$\frac{\partial}{\partial t} A_t = (*)^3 F_{A_t}$$

For $A = A_t$ in temporal gauge, A is a SD YM connection IFF A_t follows $\text{grad}_B CS = *F_B$

Similarly, A is ASD iff A_t is a flow of $-\text{grad}_B CS$.

Now that we've made sense of what it means to be a generator of the ICC, there's one technical point we must discuss.

Just like in the proof of Donaldson's theorem, the space $\mathcal{A}/\mathcal{G}$ in the 3-dimensional setting is not a manifold, since there can be connections with nontrivial stabilizer. Recall that we proved that the stabilizer of a connection is the centralizer of the connection.

This holds in the 3-d case too.

Subgroups of $SU(2)$	Centralizer
I	$SU(2)$
$-I$	$SU(2)$
S^1 or a conjugate of S^1	S^1
Subgroups of S^1 or conjugate	S^1
non-abelian	$\pm I$
$SU(2)$	$\pm I$

Lemma Flat connections with stabilizer larger than $\pm I$ correspond bijectively to abelian representations of $\pi_1(Y)$, ie, those which factor through $H_1(Y;\mathbb{Z})$.

Def ∇ is called reducible if $\mathrm{stab}(\nabla)$ is larger than $\pm I$

This is the first place we'll use that Y^3 is a $\mathbb{Z}HS^3$.

If Y is a homology sphere, there is only one reducible flat connection which corresponds to the trivial representation into $SU(2)$

Rmk if Y is a $\mathbb{Q}HS^3$, could have torsion

Computability of I breaks down based on how computable the character variety is, but also on how computible the differential is

The sing points aren't necc. isolated, but they can be post perturbation.